

AKANSHA SUBEDI

STRUCTURAL ENGINEER

Melbourne, VIC (Australian Permanent Resident)

[Portfolio website](#)

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STATEMENT

Structural Engineer with 6+ years of experience in the analysis and design of steel and reinforced concrete structures across buildings and bridge projects. Recently relocated to Melbourne as a Permanent Resident with working familiarity with Australian Standards (AS/NZS) and experienced in applying ACI, AISC, ASCE, Eurocode, Indian, Thai and Nepali standards. Skilled in Finite Element Analysis (FEA) and structural design, performance-based design, ETABS, SAP2000, and Revit.

WORK EXPERIENCES

Structural Design Engineer (*Freelance – Thailand & Nepal*)

Jan 2024 – Present

- Performed structural analysis and design of numerous projects including steel framed residential/ resorts buildings, car parks, warehouses, a sports facility with truss roofing and RC framed commercial complexes & community center in Thailand and Nepal in accordance with ACI, AISC, ASCE, Thai (DPT 1311-50), IS 456, and NBC codes.
- Designed foundations, steel connections, base plates, and anchor bolts using engineering calculations, ensuring code compliance and practical constructability.
- Prepared structural calculations, reports, drawings and design documentation while coordinating with architects, clients and contractors.

Structural Engineer (*Computers and Structures Inc. (CSI) Bangkok*)

June 2023 – Dec 2023

- Performed verification testing of bridge analysis and design software (CSiBridge-ET), Structural detailing software (CSiDetail) and RC column design software (CSiCol) validating analysis, design, and detailing functionality against engineering standards.
- Performed manual calculations of concrete bridge components and verified software results.
- Designed PSC girders, bent caps, columns using AASHTO LRFD, Caltrans, and Eurocode.
- Developed algorithms and workflow documentation for analysis and design process.
- Prepared technical reports and collaborated remotely with international QA and dev teams.

Civil Engineer (Structural Design) (*TEAM Consultants Pvt. Ltd., Nepal*)

April 2017 – Oct 2019

- Performed linear static and dynamic analysis, calculations, and seismic design of more than 15 building structures
- Prepared structural designs, drawings, calculation reports, BOQs and cost estimates for low- to mid-rise residential and commercial buildings, a commercial complex with an RC core shear wall system, and a hospital extension with an expansion joint.
- Conducted site inspections, coordinated with architects, contractors, and local authorities, and resolved construction-related technical issues.
- Collaborated with architects, MEP engineers, geotechnical engineers, and clients throughout the design process to develop practical and coordinated structural solutions.

EDUCATION

Master of Engineering (Structural Engineering)

2020 – 2022

Asian Institute of Technology, Thailand

Bachelor of Civil Engineering

2011 – 2015

Tribhuvan University, Nepal

CORE COMPETENCIES

- Structural Analysis & Design
- Steel & Reinforced Concrete Design
- Seismic & Wind Engineering
- Performance-Based Design (PBD)
- Finite Element Analysis (FEA)
- Structural Assessment
- Calculation and Design Documentation

TECHNICAL SKILLS

- Structural Analysis & Design: **ETABS, SAP2000, SAFE, PERFORM-3D, CSiBridge, STAAD.Pro, VecTor FormWorks, SpaceGass, RAPT**
- Drafting & BIM: **AutoCAD, Revit**
- Programming & Analysis: **Python, MATLAB, Microsoft Excel**

RESEARCH EXPERIENCE

Thesis in Master in Structural Engineering

2021-2022

Topic: "The Effect of Performance-Based Wind Design on Seismic Design of a Tall Building in High Seismic and High Wind Zone"

Thesis Advisor: *Dr. Naveed Anwar*

- 93 Storied building: RCC framed core wall and BRB provided for lateral force resistance
- Studied the effect of considering non-linearity in wind design on seismic ductile design of the building through performance-based approach

Student Research Assistant (*Asian Institute of Technology*)

Jan 2021– April 2022

- Contributed to the development of the textbook "Strut and Tie Method" through literature review and technical writing.
- Conducted research on Advance Concrete Structures and Tall Buildings, and developed academic articles
- Conducted software testing and technical documentation for structural engineering software, contributing to the development of CSiDetail & CSi Designer.

Intern Engineer (*HELVETAS Swiss Inter-cooperation, Nepal*)

June 2016– March 2017

- Performed complete survey of water supply system from the water source to households and designed infrastructures such as intake, reservoirs, and pipelines

Intern Engineer (*Hydro-Consult Engineering Ltd., Nepal*)

April 2016– May 2016

- Supported structural assessments of intake structures, spillways, and powerhouse buildings
- Prepared technical drawings and FEM models under supervision of senior engineers

LICENSE AND CERTIFICATE

Engineers Australia - Migration Skills Assessment (Civil Engineer – Structural)
Nepal Engineering Council - Registered Professional Engineer

August 2024
Since June 2016

PUBLICATIONS

- ["Moment Curvature Curve: Unlocking the Hidden Treasures"](#)
- ["Basic Concepts for Considering Soil-Structure Interaction"](#)